

Answer Key - Final Exam

Exercise 1 We have to prove the reversibility for Metropolis Hastings. We have two cases:

Case 1: where $i = j$, it follows: $\pi_i p_{ii} = \pi_j p_{jj}$.

Case 2: where $i \neq j$, here:

$$\pi_i p_{ij} = \pi_i q_{ij} \times \min \left\{ 1, \frac{\pi_j q_{ji}}{\pi_i q_{ij}} \right\} = \begin{cases} \pi_j q_{ji} & \text{if } \frac{\pi_j q_{ji}}{\pi_i q_{ij}} \leq 1 \\ \pi_i q_{ij} & \text{if } \frac{\pi_j q_{ji}}{\pi_i q_{ij}} > 1 \end{cases} \quad (1)$$

Similarly

$$\pi_j p_{ji} = \pi_j q_{ji} \times \min \left\{ 1, \frac{\pi_i q_{ij}}{\pi_j q_{ji}} \right\} = \begin{cases} \pi_i q_{ij} & \text{if } \frac{\pi_i q_{ij}}{\pi_j q_{ji}} > 1 \\ \pi_j q_{ji} & \text{if } \frac{\pi_i q_{ij}}{\pi_j q_{ji}} \leq 1 \end{cases} \quad (2)$$

From (1) and (2) we obtain

$$\pi_i p_{ii} = \pi_j p_{jj}, \forall i \neq j.$$

(1,5)

Exercise 2 Let $I = \int_a^b H(x) dx$

1. We can write:

$$I = \int_a^b H(x) dx = (b-a) \int_{\mathbb{R}} H(x) \frac{1}{b-a} \mathbb{I}_{[a,b]}(x) dx = (b-a) \int_{\mathbb{R}} H(x) f(x) \mathbb{I}_{[a,b]}(x) dx$$

where $f(x) = \frac{1}{b-a}$ is density function of r.v $X \sim \mathcal{U}[a, b]$, then:

$$I = (b-a) E[H(X)].$$

Hence, for $(X_i)_{i=1}^N$ iid $\sim \mathcal{U}[a, b]$ we can approximate I by the estimator:

$$I \approx \widehat{I}_N = \frac{(b-a)}{N} \sum_{i=1}^N H(x_i).$$

(1,5)

2. Since X and $(b + a - X)$ leave invariant the uniform probability on $[a, b]$, we apply the antithetic variables method, so for $(X_i)_{i=1}^N$ iid $\sim \mathcal{U}[a, b]$:

$$\begin{aligned}
 \text{Var}(\widehat{I_{2N}}) &= (b-a)^2 \frac{1}{4N^2} \text{Var} \left[\sum_{i=1}^N (H(X_i) + H(b+a-X_i)) \right] \\
 &= \frac{(b-a)^2}{4N} \text{Var}[(H(X_i) + H(b+a-X_i))] \\
 &= \frac{(b-a)^2}{4N} [\text{Var}(H(X)) + \text{Var}(H(b+a-X)) + 2\text{cov}(H(X), H(b+a-X))] \\
 &= \frac{1}{4N} [2\text{Var}(H(X)) + 2\text{cov}(H(X), H(b+a-X))] \\
 &= \frac{1}{2N} [\text{Var}(H(X)) + \text{cov}(H(X), H(b+a-X))] .
 \end{aligned}$$

Since $H(x)$ is monotonic in x , so $H(x)$ and $H(b+a-x)$ are monotonic in opposite direction, hence $\text{cov}(H(X), H(b+a-X)) \leq 0$. Therefore,

$$\text{Var}(\widehat{I_{2N}}) \leq \frac{1}{2N} [\text{Var}(H(X))]$$

In the other side $\text{Var}(\widehat{I_N}) = (b-a)^2 \frac{1}{N} \text{Var}(H(X))$.

$$\begin{aligned}
 \text{Var}(\widehat{I_{2N}}) &\leq \frac{1}{2N} [N \text{Var}(\widehat{I_N})] \\
 &\leq \frac{1}{2} \text{Var}(\widehat{I_N}) .
 \end{aligned}
 \tag{2.5}$$

Remark: We can use a second method based on the Cauchy-Schwartz inequality.

Exercise 3 Let the density function r.v X : $f(x) = \frac{\lambda}{2} e^{-\lambda|x-\theta|}$, $x \in \mathbb{R}$, $\lambda > 0$ and $\forall \theta \in \mathbb{R}$. First we calculate the cumulative function, $F(x) = \int_{-\infty}^x f(t)dt$, so

$$F(x) = \begin{cases} \frac{1}{2} e^{-\lambda(x-\theta)}, & x \leq \theta \\ 1 - \frac{1}{2} e^{-\lambda(x-\theta)}, & x > \theta \end{cases} .$$

Then, we apply the inverse transform method: let $U \in \mathcal{U}[0, 1]$, and put $u = F(x)$, we obtain

$$X = F^{-1}(U) = \begin{cases} \theta + \frac{1}{\lambda} \ln(2u), & u \in]0, \frac{1}{2}] \\ \theta - \frac{1}{\lambda} \ln(2(1-u)), & u \in]\frac{1}{2}, 1[\end{cases}.$$

Algorithm

1. Generate $U \in \mathcal{U}[0, 1]$
2. If $u \leq \frac{1}{2}$, then $X = \theta + \frac{1}{\lambda} \ln(2u)$,
else $X = \theta - \frac{1}{\lambda} \ln(2u)$.

(3)

Exercise 4 1. In the case of an $M/M/1$ model, let's assume that the station is large enough, then $S = \{0, 1, 2, \dots\}$, with

$$\begin{aligned} \lambda &= 5 \text{ cars/hour} \\ \frac{1}{\mu} &= 10 \text{ mins/car} \Rightarrow \mu = 6 \text{ cars/hour,} \end{aligned}$$

(1)

Since $\rho = \frac{\lambda}{\mu} = \frac{5}{6} < 1$, so steady state can be reached.

a) The mean number of cars in queue

$$\bar{n}_f = \frac{\rho^2}{1 - \rho} = \frac{(5/6)^2}{1 - 5/6} = \frac{25}{6} \simeq 4.17 \simeq 4 \text{ cars.}$$

b) The mean waiting time is

$$\bar{W} = \frac{\lambda}{\mu(\mu - \lambda)} = \frac{5}{6} h = 50 \text{ minutes.}$$

c) The probability of the wash station being inactive is

$$p_0 = 1 - \rho \simeq 0.17,$$

the washing station is inactive 17% of the time.

d) The mean time spent in the garage is

$$\bar{W}_s = \frac{1}{(\mu - \lambda)} = 1 \text{ hour.}$$

(1)

2. If N is the number of spaces required (queue spaces plus wash station), the model becomes $M/M/1/N$

a) - The stationary distribution of the number of customers in the system:

$$p_n = \rho^n p_0.$$

Let's determine p_0 , from the normalization equation $\sum_{n=0}^N p_n = 1$, for $\rho < 1$, we get

$$p_0 = [1 + \rho + \rho^2 + \dots + \rho^N]^{-1} = \frac{1 - \rho}{1 - \rho^{N+1}},$$

hence

$$p_n = \rho^n \left(\frac{1 - \rho}{1 - \rho^{N+1}} \right). \quad (1)$$

- The mean number of customers in the system:

$$\begin{aligned} \bar{n} &= \sum_{n=0}^N n p_n = \frac{1 - \rho}{1 - \rho^{N+1}} \sum_{n=0}^N n \rho^n = \frac{1 - \rho}{1 - \rho^{N+1}} \left(\frac{1 - \rho^{N+1}}{1 - \rho} \right)' \\ &= \frac{\rho [1 - (N+1)\rho^N + N\rho^{N+1}]}{(1 - \rho)(1 - \rho^{N+1})}. \end{aligned} \quad (1)$$

b) A driver can park inside the garage 8 times out of 10, if $\sum_{n=0}^{N-1} p_n \geq 0.8$ so:

$$\begin{aligned} \sum_{n=0}^{N-1} p_n &= \frac{1 - \rho}{1 - \rho^{N+1}} \sum_{n=0}^{N-1} \rho^n \geq 0.8 \\ \Rightarrow \frac{1 - \rho}{1 - \rho^{N+1}} \frac{1 - \rho^N}{1 - \rho} &= \frac{1 - \rho^N}{1 - \rho^{N+1}} \geq 0.8 \\ \Rightarrow \rho^N < 0.6 \Rightarrow N &\geq 2,8 \simeq 3 \text{ positions} \end{aligned} \quad (1)$$

Consequently, 3 spaces are needed for a customer to park 8 times out of 10, inside the garage.

3. For $N = 9$ (8 queueing spaces plus 1 wash station), $\rho = \frac{5}{6} = 0.8333$ and $p_n = \rho^n \left(\frac{1 - \rho}{1 - \rho^{10}} \right) = \rho^n \times 0.1988$;

- a) the number of lost customers: a customer is lost if the system is in state $N = 9$:

$$\begin{aligned}\lambda p_9 &= \lambda \rho^9 \times \frac{(1-\rho)}{1-\rho^{10}} \simeq 5 \times 0.1938 \times 0.1988 \\ &= 5 \times 0.03853 \simeq 0.1926 \text{ cars/h.}\end{aligned}$$

(1)

If the garage is open 8 hour/day, then the number of customers lost is 1.5408, i.e. 2 customers lost per day.

- b) the mean entry rate

$$\bar{\lambda} = \lambda (1 - p_9) = 5 \times (1 - 0.03853) = 5 \times 0.96147 = 4.80735.$$

(0,75)

And the mean waiting time in the queue : $\bar{W}_s = \frac{\bar{n}}{\bar{\lambda}}$, we have :

$$\begin{aligned}\bar{n} &= \frac{\rho [1 - 10\rho^9 + 9\rho^{10}]}{(1-\rho)(1-\rho^{10})} = \frac{\frac{5}{6} \left[1 - 10 \left(\frac{5}{6} \right)^9 + 9 \left(\frac{5}{6} \right)^{10} \right]}{\frac{1}{6} \left[1 - \left(\frac{5}{6} \right)^{10} \right]} \\ &= 5 \times \frac{0.5155}{0.8385} = 5 \times 0.6148 \simeq 3.0742 \text{ cars}\end{aligned}$$

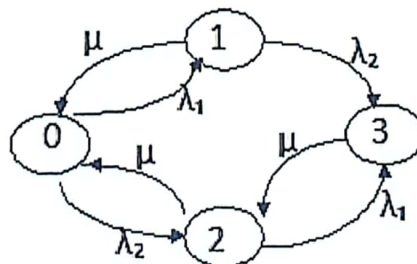
So

$$\bar{W}_s = \frac{3.0742}{4.80735} \simeq 0.6395 \text{ hour} \simeq 38 \text{ mins.}$$

(0,75)

Exercise 5 Let the Markov Chain $\{X(t), t \geq 0\}$ and $S = \{0, 1, 2, 3\}$

1. - The representative graph



(1)

- The transition matrix

$$P = \begin{pmatrix} 0 & \frac{\lambda_1}{\lambda_1 + \lambda_2} & \frac{\lambda_2}{\lambda_1 + \lambda_2} & 0 \\ \frac{\mu}{\lambda_2 + \mu} & 0 & 0 & \frac{\lambda_2}{\lambda_2 + \mu} \\ \frac{\mu}{\lambda_1 + \mu} & 0 & 0 & \frac{\lambda_1}{\lambda_1 + \mu} \\ 0 & 0 & 1 & 0 \end{pmatrix}. \quad (1)$$

2. Algorithm

1. *Initialisation:* $X_0 = 0$,

2. *Generate* $U_0 \in \mathcal{U}[0, 1]$

If $u_0 \in \left[0, \frac{\lambda_1}{\lambda_1 + \lambda_2}\right] \rightarrow$ the event $A : X_1 = 1$,

If $u_0 \in \left[\frac{\lambda_1}{\lambda_1 + \lambda_2}, 1\right] \rightarrow$ the event $B : X_1 = 2$,

3. *Generate* $U_1, U_2 \in \mathcal{U}[0, 1]$

3.1 *If* A :

if $u_1 \in \left[0, \frac{\lambda_2}{\lambda_2 + \mu}\right] \rightarrow$ next event $C_1 : X_2 = 3$, and $\tau_1 = -\frac{1}{\lambda_2} \ln(u_2)$
else $\rightarrow$ next event $C_2 : X_2 = 0$, and $\tau_1 = -\frac{1}{\mu} \ln(u_2)$

3.2 *If* B :

if $u_1 \in \left[0, \frac{\lambda_1}{\lambda_1 + \mu}\right] \rightarrow$ next event $C_1 : X_2 = 3$ and $\tau_1 = -\frac{1}{\lambda_1} \ln(u_2)$
else $\rightarrow$ next event $C_2 : X_2 = 0$ and $\tau_1 = -\frac{1}{\mu} \ln(u_2)$,

4. *Generate* $U_3 \in \mathcal{U}[0, 1]$

$$\tau_2 = -\frac{1}{\mu} \ln(u_3),$$

If $C_1, \{X_3 = 2\}$ go to 3.2,

If $C_2, \{X_3 = 0\}$ go to 2.

(2,5)